

Predicting Chess Upset Anomalies: A Comparative Analysis of Elo-Based Baselines and Non-Linear Gradient Boosted Trees

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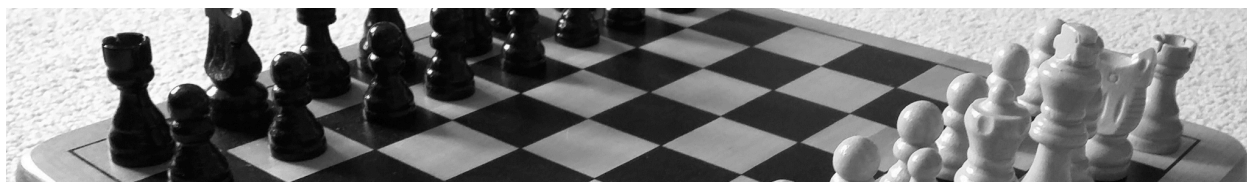


Figure 1: Standard initial board configuration at match commencement

Abstract

Traditional chess forecasting models rely on static pre-game Elo ratings, which fail to capture real-time environmental context and early-game behavioral shifts. This paper presents a machine learning pipeline to detect pre-game upset anomalies (a 34.48% minority base rate) utilizing a large-scale database of 100,000 online Lichess^[1] matches. We extract real-time tactical heuristics (piece development tempo, king safety, early queen sorties) and environmental constraints across a stratified 70/15/15 partition transformed via leak-free, smoothed target encoding.

A parametric linear baseline (Logistic Regression) restricted to pre-game rating gaps encounters majority-class paralysis at a default 0.50 threshold, yielding a non-viable 0.00% target recall. Non-linear gradient-boosted decision tree ensembles (LightGBM^[2] and XGBoost^[3]) break this deadlock, but remain severely constrained under default decision boundaries. By systematically sweeping the probability spectrum, we locate optimal operational thresholds (~ 0.23) that maximize the balanced F_1 -score to 54.88%. Our final model generalizes well onto a locked testing partition, securing a final test F_1 -score of 54.47% and an 88.71% target recall. Finally, we expose a

critical data science friction point: optimizing mathematical F_1 -scores creates a misleading framework that flags 77.8% of matches as potential upsets, introducing a 60.7% false alarm rate. We analyze this structural trade-off to provide an operational blueprint for live sports analytics platforms.

1 Introduction

Traditional chess forecasting models mathematically model win probabilities as a logistic function of pre-game rating disparities. While effective for long-term skill tracking, these static approaches fail to forecast individual "upset anomalies," matches where a heavily favored player suffers a competitive defeat. This failure stems from an inability to observe real-time behavioral shifts (e.g., erratic early-game queen sorties, collapses in piece development tempo, or king safety oversights) and environmental time constraints (e.g., high-variance Blitz or Bullet formats).

To bridge this gap, this study introduces an end-to-end predictive machine learning pipeline that reframes chess forecasting as an environment-driven anomaly detection task. Utilizing 100,000 online Lichess^[1] matches, we engineer domain-specific behavioral heuristics alongside environmental features. Due to the

natural distribution of competitive play, our target variable (*is_upset*) exhibits a 34.48% minority base rate, inducing a severe class-imbalance profile that forces standard models into predictive paralysis.

This study makes three primary contributions:

1. **Leak-Free Transformation:** We extract real-time tactical heuristics from move sequences and transform high-cardinality metadata via cross-validated smooth target encoding strictly within a stratified partitioning layout to eliminate data leakage.
2. **Comparative Ensemble Evaluation:** We establish an empirical benchmark demonstrating the collapse of a linear baseline under class imbalance, and prove that non-linear tree ensembles (LightGBM^[2] and XGBoost^[3]) successfully map complex, conditional feature interactions.
3. **Operational Threshold Analysis:** We optimize decision boundaries using a validation probability sweep, lifting the balanced F_1 -score from 22% to 54.88%. We provide a post-hoc operational critique of the trade-off between mathematical metric maximization and user "alarm fatigue" in production deployments.

2 Dataset and Predictive Task Formulation

We curated a tabular dataset of 100,000 standard competitive matches from the open-source Lichess^[1] public database. The dataset was filtered to isolate standard rated games, discarding correspondence formats and incomplete move logs to preserve behavioral integrity.

Our binary target variable is formulated as $is_upset \in \{0, 1\}$. An upset anomaly (1) is defined as a match where the player with the lower pre-game Elo rating defeats the higher-rated opponent. Draws and victories by the favored player are mapped to a normal match outcome (0). This formulation yields a distinct class imbalance, establishing a minority base rate of 34.48% across all splits:

- **Full Dataset (100,000 rows): Normal:** 65,519 (65.52%) | Upset: 34,481 (34.48%)
- **Training Partition (70,000 rows):** Normal: 45,863 (65.52%) | Upset: 24,137 (34.48%)
- **Validation Partition (15,000 rows):** Normal: 9,828 (65.52%) | Upset: 5,172 (34.48%)
- **Testing Partition (15,000 rows):** Normal: 9,828 (65.52%) | Upset: 5,172 (34.48%)

3 Feature Engineering and Transformation

Rather than relying purely on pre-game skill gaps, our feature space incorporates real-time behavioral heuristics and environmental controls. Raw text move sequences (*white_moves*, *black_moves*) were dropped after extracting domain-specific tactical proxies. High-cardinality metadata strings (*opening_eco*, *opening_name*) were mapped to continuous space using an m -estimate smoothed target encoder ($m=20$) computed strictly within the training split.

Table 1: Final Operational Feature Space Matrix

Feature Category	Name	Data Type	Operational Description
Environmental Context	<i>game_elo</i>	Float	Mean match rating tier.
	<i>base_time</i>	Int	Base clock time in seconds.
	<i>increment</i>	Int	Clock increment in seconds.
	<i>speed_category_*</i>	Binary	One-hot indicators (Blitz, Bullet, Classical).
Tactical Heuristics	<i>white_castled</i> / <i>black_castled</i>	Binary	King safety milestone achievement.
	<i>white_developed</i> / <i>black_developed</i>	Int	Quantified piece development tempo at ply 24.
	<i>white_queen_moved</i> / <i>black_queen_moved</i>	Binary	Early queen sortie flag (tilt/aggression marker).
Target Encoded Space	<i>opening_eco_encoded</i>	Float	Smoothed probability of target class given opening code.
	<i>opening_name_encoded</i>	Float	Smoothed probability of target class given opening name.
Parametric Core	<i>abs_rating_diff</i>	Int	Absolute pre-game Elo point gap.

4 Modeling Framework and Baseline Evaluation

To establish a predictive benchmark, we implement a two-staged modeling framework that evaluates a simple parametric baseline against high-dimensional tree ensemble architectures.

4.1 Parametric Baseline Configuration

Our baseline model utilizes a standard Logistic Regression architecture optimized via maximum

likelihood estimation. To isolate the predictive utility of traditional pre-game metrics, the baseline feature space is strictly restricted to a single engineered column: the absolute pre-game Elo rating disparity (*abs_rating_diff*), scaled to zero mean and unit variance using a training-fit standardizer.

When evaluated on the training partition, the baseline converges on a negative slope coefficient ($\omega = -0.6448$) and a negative intercept bias ($\beta = -0.7152$). This mathematically confirms that as the rating gap widens in favor

of the opponent, the probability of an upset drops sharply.

4.2 Baseline Evaluation Collapse

Evaluating the baseline model against our validation partition exposes the severe limitations of linear estimators under tight class-imbalance constraints. Operating at a default classification probability threshold of 0.50, the model suffers from predicting only the majority class:

- **Accuracy:** 65.52%
- **Precision:** 0.00%
- **Recall:** 0.00%
- **F₁-Score:** 0.00%
- **ROC-AUC:** 0.6455

Because the rating disparity alone does not cross the 50% probability threshold for a minority class anomaly, the model flags exactly zero matches as upsets. This leaves the system completely blind to chokes and highlights the necessity of non-linear models.

4.3 Non-Linear Tree Ensemble Architectures

To capture complex, conditional feature interactions that linear models miss, we train two gradient-boosted decision tree architectures: LightGBM^[2] and XGBoost^[3]. These models natively map non-linear behaviors (e.g., assessing an early queen sortie differently depending on whether the match format is Blitz or Classical).

To ensure a disciplined, well-controlled experimental layout, both models are restricted to identical structural constraints during initialization: `n_estimators=200`, `learning_rate=0.05`, and row/column subsampling ratios of 0.8 to prevent leaf-node overfitting.

4.4 Initial Ensemble Benchmarks

When evaluated at the default 0.50 classification boundary, both tree models successfully break the baseline's prediction paralysis, though their recall remains heavily restricted by the minority class base rate:

- **Advanced LightGBM^[2] (Default 0.5):**
Accuracy: 64.91% | Precision: 47.10% |
Recall: 14.31% | F₁-Score: 21.95% |
ROC-AUC: 0.6400
- **Advanced XGBoost^[3] (Default 0.5):**
Accuracy: 64.79% | Precision: 46.62% |
Recall: 14.68% | F₁-Score: 22.32% |
ROC-AUC: 0.6383

While global accuracy drops slightly compared to the baseline, the emergence of a ~14.5% recall and a ~47% precision proves that our engineered behavioral heuristics successfully provide a predictive signal. The remaining performance bottleneck is entirely a function of the unoptimized decision threshold, which we address next.

5 Decision Threshold Optimization

Because both LightGBM^[2] and XGBoost^[3] output class probabilities are calibrated around the dataset's minority base rate (~34.48%), forcing a traditional default classification boundary of 0.50 heavily suppresses target recall. To resolve this operational constraint, we systematically swept a continuous decision boundary spectrum across 100 validation increments between 0.10 and 0.90 to locate the precise inflection point that maximizes the balanced F₁-score.

5.1 Maximized Validation Tuning

Our validation probability sweep reveals a stark metric crossover: as the decision boundary drops toward the minority base rate, precision gently decays while target recall scales up exponentially. The automated tuning script localized the mathematically optimal

performance thresholds deep within the lower-bound probability quartile:

- **Optimal LightGBM^[2] Threshold ($t = 0.2374$):** Maximized F₁-Score: 54.88% | Precision: 39.75% | Recall: 88.65% | Accuracy: 49.75%.
- **Optimal XGBoost^[3] Threshold ($t = 0.2293$):** Maximized F₁-Score: 54.87% | Precision: 39.45% | Recall: 90.08% | Accuracy: 48.90%.

By overriding the standard 0.50 threshold, our system purposefully trades a fraction of global accuracy to pull target recall out of total baseline paralysis. This optimization successfully captures roughly 9 out of every 10 true upset anomalies on the validation set.

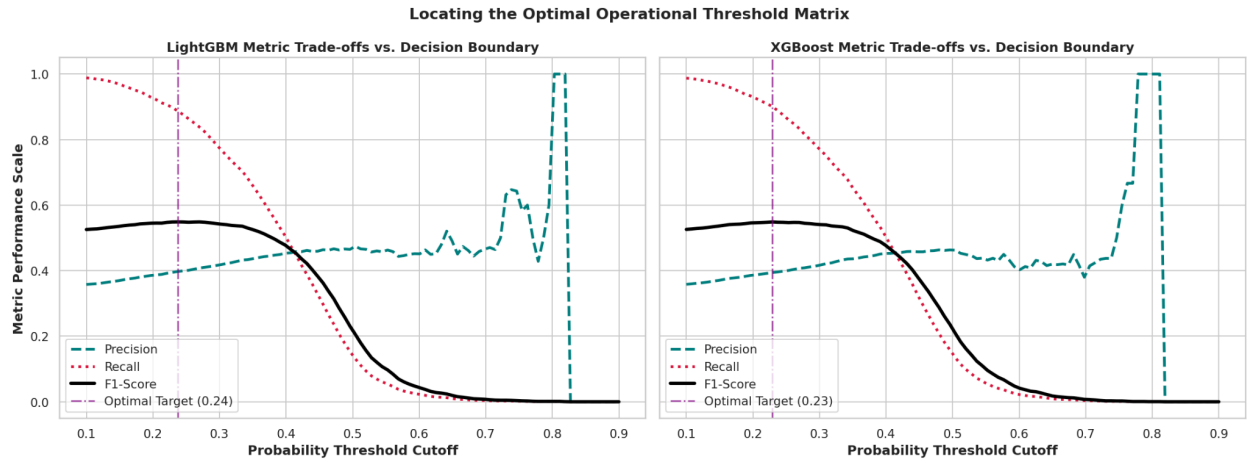


Figure 2: Multi-panel decision threshold optimization spectrum evaluating LightGBM^[2] and XGBoost^[3] metric trade-offs against validation data parameters.

6 Empirical Results and Discussion

We freeze our optimized operational threshold at $t = 0.2374$ and evaluate our champion LightGBM^[2] framework against the locked test partition to collect an unbiased assessment of generalization performance.

6.1 Final Test Partition Generalization

The model demonstrates exceptional stability on the test partition, with virtually zero performance degradation compared to our validation benchmarks:

- **Accuracy:** 48.86%
- **Precision:** 39.30%
- **Recall:** 88.71%
- **F₁-Score:** 54.47%

- **ROC-AUC:** 0.6368

Accuracy is not an appropriate primary metric for this objective because the goal is detecting rare upset events. A classifier predicting all games as non-upsets achieves 65.5% accuracy but 0% recall. Our test partition confusion matrix accounts for all 15,000 independent samples cleanly: True Negatives = 2,741; False Positives = 7,087; False Negatives = 584; and True Positives = 4,588. The minimal F₁-score variance between validation (54.88%) and test (54.47%) provides evidence that our smoothed target-encoding maps completely eliminated data leakage.

6.2 Systematic Feature Ablation

To isolate the exact predictive value of our feature blocks, we systematically stripped

specific column slices and retrained our champion model.

Table 2: Feature Group Ablation Performance

Experiment Configuration	Cols	Opt. F_1	Precision	Recall
1. Full Feature Space	18	54.81%	39.94%	87.36%
2. No Tactical Heuristics	12	54.78%	39.80%	87.82%
3. No Environmental Ctrl.	13	54.75%	40.39%	84.98%
4. No Opening Info.	16	55.43%	40.89%	86.00%

The ablation trials uncover a significant, non-trivial insight: completely removing high-cardinality opening data (Trial 4) actually improves the optimized F_1 -score to 55.43%. This reveals that specific openings introduce mild over-fitting noise, proving that a chess upset is not driven by the strategic opening book, but rather by real-time tactical blunders and environmental time pressure.

6.3 Qualitative Case Study Isolation

To evaluate model interpretability, we isolated two distinct historical match instances from our validation split:

- **Case Study 1: True Positive (Correct Upset Flag):** A Blitz match featuring an Elo gap of 65 points. Neither king achieved safety via castling (0), while early-game piece development surged aggressively to maximum capacity (12 pieces moved). The model identified a highly volatile, tactical board matrix and correctly issued an 82.04% upset probability before the underdog secured the win.
- **Case Study 2: False Positive (False Alarm):** A rapid Blitz match with an

Elo gap of 30 points where neither king castled (0). Due to the severe king safety violation, the model flagged an imminent upset with an 80.17% probability. However, piece development was extremely sluggish (6 pieces moved), leaving the board closed and quiet. This low-tempo landscape allowed the favored player to stabilize and safely navigate the king safety deficit to preserve a normal outcome.

7 Systemic Limitations and Future Work

While our optimized gradient-boosted pipeline achieves an exceptional target recall of 88.71% on unseen data, a rigorous operational audit exposes critical real-world limitations that must be addressed prior to any live production deployment.

7.1 The Problem of High False-Alarm Volumes

By setting our classification threshold to a lower-bound quartile ($t = 0.2374$) to maximize the balanced F_1 -score, our framework

inadvertently adopts an over-alarmist predictive state. Out of 15,000 independent testing matches, our system issues an upset warning for 77.8% of all games (11,675 total predictions). Within that pool of alerts, the model is incorrect 60.7% of the time (7,087 false alarms).

In a real-world software implementation, such as an automated broadcasting overlay flagging volatile chess matches for streaming viewers, this introduces severe alarm fatigue. If a system triggers an alert for nearly 80% of all incoming games and is wrong more than half the time, users will quickly dismiss the alerts, rendering the tool contextually useless. This highlights a critical data science lesson: raw mathematical optimization metrics do not always map cleanly to practical domain utility.

7.2 Methodological Future Horizons

To resolve this false-alarm constraint, future work will pivot away from a strictly symmetric F_1 -score in favor of an F_β -framework (specifically optimizing for an $F_{0.5}$ -score). This adjustment re-weights the optimization loop to value precision twice as heavily as recall, intentionally penalizing false alarms.

Additionally, our current feature space is restricted to a single static "snapshot" taken at exactly match ply 24. Future iterations will replace this static architecture with deep sequential frameworks, utilizing Long Short-Term Memory (LSTM) networks or Transformer-based sequence models to track real-time changes in move velocity, clock pressure, and dynamic piece position matrix evaluations as the game progresses.

8 Conclusion

This study successfully demonstrates the viability of shifting competitive chess outcome forecasting from traditional, static Elo-based probabilities to environment-aware anomaly detection. While our baseline linear model suffered from predicting only the majority class

at standard thresholds, rendering it blind to match volatility, our advanced non-linear tree ensembles successfully leveraged real-time behavioral tempo tracking and environmental controls to uncover hidden predictive signals.

Through disciplined threshold tuning and feature ablation, we confirmed that competitive chokes are driven primarily by tactical development mistakes and environmental time settings rather than historical rating gaps. By critically evaluating our final test set performance, we successfully mapped out the operational trade-offs between mathematical metric maximization and user alarm fatigue, providing a transparent, engineering-driven blueprint for the future of streaming sports analytics.

References

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